Assignment - 4 (DA)

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```r
# Load required libraries
library(MASS)

## Warning: package 'MASS' was built under R version 4.4.3

# Load the Wine dataset
wine <- read.table("http://archive.ics.uci.edu/ml/machine-learning-
databases/wine/wine.data",
                   sep=",")

# Add column names
colnames(wine) <- c("Type", "Alcohol", "Malic_acid", "Ash",
                    "Alcalinity_of_ash", "Magnesium", "Total_phenols",
                    "Flavanoids", "Nonflavanoid_phenols", "Proanthocyanins",
                    "Color_intensity", "Hue", "OD280_OD315", "Proline")

# Convert Type to factor
wine$Type <- as.factor(wine$Type)

# ====================================================================
# DATA EXPLORATION
# ====================================================================

# View first few rows
head(wine)

##   Type Alcohol Malic_acid  Ash Alcalinity_of_ash Magnesium Total_phenols
## 1    1   14.23       1.71 2.43              15.6       127          2.80
## 2    1   13.20       1.78 2.14              11.2       100          2.65
## 3    1   13.16       2.36 2.67              18.6       101          2.80
## 4    1   14.37       1.95 2.50              16.8       113          3.85
## 5    1   13.24       2.59 2.87              21.0       118          2.80
## 6    1   14.20       1.76 2.45              15.2       112          3.27
##   Flavanoids Nonflavanoid_phenols Proanthocyanins Color_intensity  Hue
## 1       3.06                 0.28            2.29            5.64 1.04
## 2       2.76                 0.26            1.28            4.38 1.05
## 3       3.24                 0.30            2.81            5.68 1.03
## 4       3.49                 0.24            2.18            7.80 0.86
## 5       2.69                 0.39            1.82            4.32 1.04
## 6       3.39                 0.34            1.97            6.75 1.05
##   OD280_OD315 Proline
## 1        3.92    1065
## 2        3.40    1050
## 3        3.17    1185
```

```
## 4        3.45    1480
## 5        2.93     735
## 6        2.85    1450

# Check dimensions
dim(wine)

## [1] 178  14

# Check class distribution
table(wine$Type)

##
##  1  2  3
## 59 71 48

# Summary statistics
summary(wine)

##  Type      Alcohol        Malic_acid         Ash        Alcalinity_of_ash
##  1:59   Min.   :11.03   Min.   :0.740   Min.   :1.360   Min.   :10.60
##  2:71   1st Qu.:12.36   1st Qu.:1.603   1st Qu.:2.210   1st Qu.:17.20
##  3:48   Median :13.05   Median :1.865   Median :2.360   Median :19.50
##         Mean   :13.00   Mean   :2.336   Mean   :2.367   Mean   :19.49
##         3rd Qu.:13.68   3rd Qu.:3.083   3rd Qu.:2.558   3rd Qu.:21.50
##         Max.   :14.83   Max.   :5.800   Max.   :3.230   Max.   :30.00
##    Magnesium      Total_phenols     Flavanoids    Nonflavanoid_phenols
##  Min.   : 70.00   Min.   :0.980   Min.   :0.340   Min.   :0.1300
##  1st Qu.: 88.00   1st Qu.:1.742   1st Qu.:1.205   1st Qu.:0.2700
##  Median : 98.00   Median :2.355   Median :2.135   Median :0.3400
##  Mean   : 99.74   Mean   :2.295   Mean   :2.029   Mean   :0.3619
##  3rd Qu.:107.00   3rd Qu.:2.800   3rd Qu.:2.875   3rd Qu.:0.4375
##  Max.   :162.00   Max.   :3.880   Max.   :5.080   Max.   :0.6600
##  Proanthocyanins Color_intensity       Hue           OD280_OD315
##  Min.   :0.410   Min.   : 1.280   Min.   :0.4800   Min.   :1.270
##  1st Qu.:1.250   1st Qu.: 3.220   1st Qu.:0.7825   1st Qu.:1.938
##  Median :1.555   Median : 4.690   Median :0.9650   Median :2.780
##  Mean   :1.591   Mean   : 5.058   Mean   :0.9574   Mean   :2.612
##  3rd Qu.:1.950   3rd Qu.: 6.200   3rd Qu.:1.1200   3rd Qu.:3.170
##  Max.   :3.580   Max.   :13.000   Max.   :1.7100   Max.   :4.000
##     Proline
##  Min.   : 278.0
##  1st Qu.: 500.5
##  Median : 673.5
##  Mean   : 746.9
##  3rd Qu.: 985.0
##  Max.   :1680.0
```

Wine Dataset Summary

Dataset Structure

Size: 178 wines × 14 variables (13 chemical features + 1 type) Classes: Type 1 (59), Type 2 (71), Type 3 (48) Balance: Fairly balanced dataset

Key Features Range

Alcohol: 11.03% to 14.83% (mean: 13.00%) Proline: Wide range (278-1680), high variability Flavanoids: 0.34 to 5.08, important discriminator Color Intensity: 1.28 to 13.00, varies significantly

```
# ======================================================================
# FIT LDA MODEL
# ======================================================================

# Build LDA model
model_lda <- lda(Type ~ ., data = wine)

# Display model summary
model_lda

## Call:
## lda(Type ~ ., data = wine)
##
## Prior probabilities of groups:
##         1         2         3
## 0.3314607 0.3988764 0.2696629
##
## Group means:
##    Alcohol Malic_acid      Ash Alcalinity_of_ash Magnesium Total_phenols
## 1 13.74475   2.010678 2.455593          17.03729  106.3390      2.840169
## 2 12.27873   1.932676 2.244789          20.23803   94.5493      2.258873
## 3 13.15375   3.333750 2.437083          21.41667   99.3125      1.678750
##   Flavanoids Nonflavanoid_phenols Proanthocyanins Color_intensity
Hue
## 1  2.9823729             0.290000        1.899322        5.528305
1.0620339
## 2  2.0808451             0.363662        1.630282        3.086620
1.0562817
## 3  0.7814583             0.447500        1.153542        7.396250
0.6827083
##   OD280_OD315   Proline
## 1    3.157797 1115.7119
## 2    2.785352  519.5070
## 3    1.683542  629.8958
##
```

```
## Coefficients of linear discriminants:
##                                LD1           LD2
## Alcohol              -0.403399781  0.8717930699
## Malic_acid            0.165254596  0.3053797325
## Ash                  -0.369075256  2.3458497486
## Alcalinity_of_ash     0.154797889 -0.1463807654
## Magnesium            -0.002163496 -0.0004627565
## Total_phenols         0.618052068 -0.0322128171
## Flavanoids           -1.661191235 -0.4919980543
## Nonflavanoid_phenols -1.495818440 -1.6309537953
## Proanthocyanins       0.134092628 -0.3070875776
## Color_intensity       0.355055710  0.2532306865
## Hue                  -0.818036073 -1.5156344987
## OD280_OD315          -1.157559376  0.0511839665
## Proline              -0.002691206  0.0028529846
##
## Proportion of trace:
##    LD1    LD2
## 0.6875 0.3125
```

LDA Interpretation

Class Distribution

Type 1: 33% | Type 2: 40% | Type 3: 27%

Main Differences

- **Type 1**: High Proline (1116), high Flavanoids (2.98)
- **Type 2**: Low alcohol (12.28), low Proline (520)
- **Type 3**: Low Flavanoids (0.78), high color intensity (7.40)

Key Features

Most Important: Flavanoids (-1.66), Nonflavanoid_phenols (-1.50), OD280_OD315 (-1.16)
Least Important: Magnesium, Proline

Discriminant Functions

- **LD1**: 68.75% variance - main separator
- **LD2**: 31.25% variance - refines classification

```r
# ====================================================================
# PREDICTIONS
# ====================================================================

# Predict on the same dataset
predictions <- predict(model_lda, wine)

# View predicted classes
predictions$class
```

```
##   [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
1 1 1
##  [38] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2
2 2 2
##  [75] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
2 2 2
## [112] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
3 3 3
## [149] 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
## Levels: 1 2 3
```

```r
# View first 20 predictions
predictions$class[1:20]
```

```
##  [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## Levels: 1 2 3
```

```r
# ====================================================================
# CONFUSION MATRIX AND ACCURACY
# ====================================================================

# Create confusion matrix
confusion_matrix <- table(Predicted = predictions$class, Actual = wine$Type)
confusion_matrix
```

```
##          Actual
## Predicted  1  2  3
##         1 59  0  0
##         2  0 71  0
##         3  0  0 48
```

```r
# Calculate overall accuracy
accuracy <- sum(diag(confusion_matrix)) / sum(confusion_matrix)
accuracy
```

```
## [1] 1

# Display as percentage
cat("Overall Accuracy:", round(accuracy * 100, 2), "%\n")

## Overall Accuracy: 100 %
```

Signs of Overfitting:

#100% Training Accuracy - This is suspiciously perfect

#No Train-Test Split - We’re testing on the same data we trained on

#Small Dataset - Only 178 observations with 13 features

#High Dimensionality - 13 predictors for 178 samples means high risk